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An introduction to MCMC and Bayesian Inference

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1 Discrete Markov chains

Definition 1. A *discrete Markov chain of first order* is a stochastic process $\{X_t\}_t$, $t = 1, 2, 3, \dots$ with the discrete state space S , whose *transition probabilities* fulfill the *Markov prop-*

erty

$$P(X_t = x_t | X_{t-1} = x_{t-1}, \dots, X_1 = x_1) = P(X_t = x_t | X_{t-1} = x_{t-1}), \quad (1)$$

for all states $x_1, \dots, x_t \in S$. A discrete Markov chain is called **homogeneous** if the transition probabilities do not depend on t , that is

$$T(x, y) := P(X_t = y | X_{t-1} = x) = P(X_2 = y | X_1 = x), \quad (2)$$

which is then the 1-step **transition kernel** of the homogeneous (discrete) 1st-order Markov chain.

In the following, we assume that the state space S is given by a finite set. Furthermore, we assume, without loss of generality, that the state space is given by $S = \{1, 2, 3, \dots, k\}$, where k is the number of possible states, or the **cardinality** of S . This means that the transition kernel $T(\cdot, \cdot)$ can be represented as a $k \times k$ matrix, called the **transition matrix**, whose (i, j) -th element is defined as

$$T_{ij} := T(i, j) = P(X_{t+1} = j | X_t = i) = P(X_2 = j | X_1 = i). \quad (3)$$

The **initial distribution** $P(X_1 = x)$, $x \in S$, and the transition matrix T together fully define the distribution of the homogeneous Markov chain.

Definition 2. A **stochastic matrix** is a square matrix whose row or column sums are equal to one, and whose elements take values between zero and one.

Note 1. It is worth noting that the transition matrices of discrete Markov chains (with finite state space) are stochastic matrices, as each row-sum is equal to 1.

Example 1. Let's consider a Markov chain with $k = 4$ states, $S = \{1, 2, 3, 4\}$, initial distribution $P(X_1 = i) = 0.25$, $i = 1, \dots, 4$, and transition matrix

$$T = \begin{bmatrix} 0.5 & 0.45 & 0.05 & 0 \\ 0.48 & 0.5 & 0 & 0.02 \\ 0.01 & 0 & 0.99 & 0 \\ 0 & 0.001 & 0 & 0.999 \end{bmatrix} \quad (4)$$

Then, the distribution at the next two time points can be computed iteratively as follows

$$\begin{aligned}
P(X_2 = j) &= \sum_{i=1}^4 P(X_1 = i)T_{ij} \\
P(X_3 = j) &= \sum_{i=1}^4 P(X_2 = i)T_{ij} \\
&= \sum_{i=1}^4 \left(\sum_{k=1}^4 P(X_1 = k)T_{ki} \right) T_{ij} \\
&= \sum_{k=1}^4 P(X_1 = k) \sum_{i=1}^4 T_{ki}T_{ij} \\
&= \sum_{k=1}^4 P(X_1 = k)(TT)_{kj} \\
&= \sum_{k=1}^4 P(X_1 = k)T_{kj}^2 \\
&= \sum_{i=1}^4 P(X_1 = i)T_{ij}^2.
\end{aligned} \tag{5}$$

This result can be generalised as follows

Proposition 1. For discrete homogeneous Markov chains with finite state space $S = \{1, 2, \dots, k\}$, an initial distribution $P(X_1 = i) = p_{1i}$, $i = 1, \dots, k$, and k -by- k transition matrix T , we have

$$P(X_t = j) = \sum_{i=1}^k p_{1i} T_{ij}^{t-1}, \tag{6}$$

and thus

$$\begin{aligned}
(p_{t1}, \dots, p_{tk}) &= (p_{11}, \dots, p_{1k})T^{t-1} \\
\mathbf{p}_t &= \mathbf{p}_1 T^{t-1},
\end{aligned} \tag{7}$$

where

$$\mathbf{p}_t = (p_{t1}, \dots, p_{tk}). \tag{8}$$

That is, the $(t-1)$ -step transition probabilities $\mathbf{p}_t$ at time t can be obtained by multiplying the initial distribution $\mathbf{p}_1$ with T^{t-1} , the transition matrix raised to the $(t-1)$ power.

Example 2. Compute the values of $\mathbf{p}_t$, for $t = 1, 2, 3, 4, 8, 16, 32, 64, 128, 256, 512, 2024, 4048, 9096$, for the Markov chain of previous example.

Solution:

```
T <- matrix(data = c(0.5,0.45,0.05,0,
                     0.48,0.5,0,0.02,
                     0.01,0,0.99,0,
```

```

      0,0.001,0,0.999),
      byrow = TRUE,
      nrow = 4,
      ncol = 4)

p0 <- rep(0.25,4)

tps <- c(1,2,3,4,8,16,32,64,128,256,512,2024,4048,9096)

res <- t(sapply(tps, function(k){
  Tk <- T
  if(k > 1){
    for (i in 1:(k-1)) {
      Tk <- Tk %*% T
    }
  }
  return(p0 %*% Tk)
}))
rownames(res) <- tps

xtable::xtable(res)

```

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_1	0.2475	0.2377	0.2600	0.2547
p_2	0.2405	0.2305	0.2698	0.2593
p_3	0.2336	0.2237	0.2791	0.2636
p_4	0.2270	0.2172	0.2880	0.2678
p_8	0.2032	0.1939	0.3195	0.2834
p_{16}	0.1663	0.1577	0.3666	0.3093
p_{32}	0.1214	0.1137	0.4177	0.3472
p_{64}	0.0859	0.0792	0.4399	0.3950
p_{128}	0.0699	0.0640	0.4093	0.4568
p_{256}	0.0596	0.0549	0.3420	0.5435
p_{512}	0.0486	0.0451	0.2634	0.6429
p_{2024}	0.0390	0.0366	0.1954	0.7290
p_{4048}	0.0389	0.0365	0.1946	0.7299
p_{9096}	0.0389	0.0365	0.1946	0.7299

Note that by repeating the same exercise with a different initial probability distribution $\mathbf{p}_1$ will lead to the same distribution at the final iteration. We will refer later to this distribution as the stationary distribution of the Markov chain. $\square$

Algorithm 1. A concrete trajectory (realization) of a given Markov chain can be generated via the following two-step iterative procedure.

1. (**Initialization**): Draw a realization x_1 of the random variable X_1 from the initial distribution $\mathbf{p}_1$.

2. (**Iteration**): For $t = 2, \dots, T$, given the realization $x_{t-1} = i$ of X_{t-1} , draw a realization of the next random variable X_t from $P(X_t|X_{t-1} = i)$. This conditional distribution is given by

$$P(X_t = j|X_{t-1} = i) = T_{ij}, \quad (9)$$

that is the i -th row of the transition matrix T .

After k iterations, it is also possible to compute the relative frequencies h_{ki} , defining how often the J states have been realized.

Example 3. Let's consider again a Markov chain with $k = 4$ states, $S = \{1, 2, 3, 4\}$, initial distribution $P(X_1 = i) = 0.25$, $i = 1, \dots, 4$, and transition matrix

$$T = \begin{bmatrix} 0.5 & 0.45 & 0.05 & 0 \\ 0.48 & 0.5 & 0 & 0.02 \\ 0.01 & 0 & 0.99 & 0 \\ 0 & 0.001 & 0 & 0.999 \end{bmatrix}. \quad (10)$$

Simulate a concrete trajectory (realization) of this Markov chain. Generate realizations of the first 10000 random variables, and report the relative frequencies h_{ki} , every 1000 steps.

Solution:

```
nIT <- 10000
x <- rep(NA, nIT)
x_freq <- matrix(data = 0, nrow = 10, ncol = 4)
colnames(x_freq) <- 1:4
x[1] <- sample(x = 1:4, size = 1, prob = p0)
for (k in 2:nIT) {
  x[k] <- sample(x = 1:4, size = 1, prob = T[x[k-1],])

  if(k %% 1000 == 0){
    x_freq[k/1000, names(table(x[1:k]))] <- table(x[1:k])/k
  }
}
xtable::xtable(x_freq, digits = 4)
```

	1	2	3	4
1	0.0050	0.0140	0.4150	0.5660
2	0.0025	0.0070	0.2075	0.7830
3	0.0077	0.0130	0.1417	0.8377
4	0.0150	0.0215	0.1095	0.8540
5	0.0204	0.0266	0.1292	0.8238
6	0.0262	0.0293	0.2225	0.7220
7	0.0296	0.0313	0.2293	0.7099
8	0.0348	0.0362	0.2160	0.7130
9	0.0309	0.0322	0.1920	0.7449
10	0.0319	0.0319	0.1728	0.7634

Note that the values of last row of this table are comparable to those of the distribution at last iteration obtained in the previous exercise. $\square$

1.1 Stationary distribution of a discrete Markov chain

Any discrete distribution π over a set $S = \{1, \dots, k\}$ is fully specified by its discrete density function. Let p_i be the probability of the realization of the value $i \in \{1, \dots, k\}$. We further assume that $p_i > 0$ for $i = 1, \dots, k$. That is, we assume that each $i \in S$ has a positive probability. Note that each $i \in S$ with $p_i = 0$ can be simply excluded from S . Then, the discrete distribution π can be identified with the row vector $\pi = (p_1, \dots, p_k)$.

Definition 3. A discrete distribution $\pi = (p_1, \dots, p_k)$ over a set $S = \{1, \dots, k\}$ is called a **stationary distribution** of a discrete Markov chain with transition matrix T if

$$\pi T = \pi. \quad (11)$$

Note 2. The stationary distribution of a Markov chain does not depend on the initial distribution, and it is fully specified by the transition matrix T .

1.2 Aperiodic Markov chains

Definition 4. A discrete Markov chain defined over a finite state space S is called **periodic** if it can reach some states of the state space S only in a certain order, otherwise the Markov chain is called **aperiodic**.

Note 3. A discrete Markov chain with finite state space S is aperiodic if for each state $i \in S$ there is a natural number $n_i \in \mathbb{N}$ such that

$$P(X_t = i | X_1 = i) > 0 \quad \forall t \geq n_i, \quad (12)$$

that is, for every state $i \in S$ the return to this state can occur in irregular time intervals, where the return may take place not earlier than after n_i steps (or transitions). This means that a Markov chain with finite state space S is periodic if there is at least one state $i \in S$ and a natural number $n_i \geq 2$, such that

$$\begin{aligned} P(X_t = i | X_1 = i) &> 0 \quad \text{for } t = n_i, 2n_i, 3n_i, \dots \\ P(X_t = i | X_1 = i) &= 0 \quad \text{for all other } t. \end{aligned} \quad (13)$$

1.3 Irreducible and ergodic Markov chains

Definition 5. A discrete Markov chain with finite state space S is called **irreducible** if every state can be reached from every other state in a finite number of steps with positive probability, i.e. if

$$\exists t > 0 \quad \text{such that } P(X_t = j | X_1 = i) > 0 \quad \forall i, j \in S. \quad (14)$$

Definition 6. A discrete Markov chain with finite state space S is called **ergodic** if it is aperiodic and irreducible.

Proposition 2. For a discrete ergodic Markov chain with finite state space S and transition matrix T , there is a unique stationary distribution π . The stationary distribution π is fully determined by the transition matrix T and it holds that

$$\pi = \mathbf{p}_1 T^t \quad \text{for } t \rightarrow \infty, \quad (15)$$

where $\mathbf{p}_1$ is an arbitrary initial distribution.

Note 4. The previous propositions implies that each discrete ergodic Markov chain with finite state space S and transition matrix T “converges” to a unique stationary distribution π , and this stationary distribution depends only on the transition matrix T .

Proposition 3. Given a discrete ergodic Markov chain with finite state space $S = \{1, \dots, k\}$ and transition matrix T , if a distribution $\pi = (p_1, \dots, p_k)$, with $p_i > 0$, fulfills for all $i, j \in S$ the so-called **equation of detailed balance**

$$\frac{T_{ij}}{T_{ji}} = \frac{p_j}{p_i}, \quad (16)$$

then the distribution π is the stationary distribution of this Markov chain.

Proof. (sketch of)

$$\begin{aligned} \frac{T_{ij}}{T_{ji}} &= \frac{p_j}{p_i} \\ T_{ij} \cdot p_i &= T_{ji} \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= \sum_{i=1}^k T_{ji} \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= \sum_{i=1}^k P(X_2 = i | X_1 = j) \cdot p_j \\ \sum_{i=1}^k T_{ij} \cdot p_i &= p_j \underbrace{\sum_{i=1}^k P(X_2 = i | X_1 = j)}_{=1} \\ \sum_{i=1}^k T_{ij} \cdot p_i &= p_j \\ \pi T &= \pi. \end{aligned} \quad (17)$$

□

Note 5. The equation (16) of detailed balance is always satisfied for $i = j$, as we have for $T_{ii} > 0$

$$1 = \frac{T_{ii}}{T_{ii}} = \frac{p_i}{p_i}, \quad (18)$$

and also note that, by definition, the equation (16) of detailed balance does not have to be satisfied if $T_{ii} = 0$.

2 Markov Chain Monte Carlo (MCMC)

2.1 Motivation

Given a discrete distribution $\pi = (p_1, \dots, p_k)$, construct an ergodic discrete Markov chain with state space $S = \{1, \dots, k\}$ and transition matrix T , so that the equation of detailed balance is

fulfilled. The distribution π is the stationary distribution of this Markov chain. That is, the so constructed Markov chain “converges” to π in the sense that if this Markov chain is simulated for “long enough”, it generates a random sample from the distribution π .

2.2 Metropolis-Hastings method

With the so-called “Metropolis-Hastings Markov Chain Monte Carlo” simulation method such a Markov chain can be constructed as follows:

For iteration $t = 1, 2, 3, \dots$

- Given the state $x_t = i$ of the Markov chain, a new state $j \in S$ is proposed with a **proposal probability** $Q(i, j)$.
- The new proposed state j is either accepted with **acceptance probability** $A(i, j)$, or rejected with the counter probability $1 - A(i, j)$.
- If the new state is accepted, set $x_{t+1} = j$, otherwise leave the state unchanged, that is, set $x_{t+1} = i$.

Let’s consider the **transition kernel** defined as

$$T(i, j) = T_{ij}, \quad (19)$$

where T_{ij} is the ij -th entry of the transition matrix T . The transition probabilities $T(i, j)$ from state i to state j have to be constructed with the so called **Metropolis-Hastings** method, that is

$$T(i, j) = \begin{cases} Q(i, j)A(i, j) & \text{if } j \neq i \\ Q(i, i)A(i, i) + 1 - \sum_{k \neq i} Q(i, k)A(i, k) & \text{if } j = i. \end{cases} \quad (20)$$

The diagonal elements of T are given by

$$T(i, i) = Q(i, i)A(i, i) + 1 - \sum_{j \neq i} Q(i, j)A(i, j), \quad (21)$$

although transitions from i to i are usually not explicitly proposed and, in fact, $Q(i, i)$ are often set to zero, so that $T(i, i)$ reduces to

$$T(i, i) = 1 - \sum_{j \neq i} Q(i, j)A(i, j). \quad (22)$$

In this case $T(i, i)$ is then simply the probability that a new proposed state $j \neq i$ is not accepted, so that the state of the Markov chain remains unchanged.

Note 6. In the Metropolis-Hastings MCMC method:

- The proposal probabilities $Q(i, j)$ are chosen arbitrarily, as long as the following two conditions are fulfilled:
 1. each state can be reached from each state in a finite number of steps.
 2. for $Q(i, j) > 0$ it follows that $Q(j, i) > 0$ too.

- Given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ have to be chosen so that the Equation (16) of detailed balance is satisfied.

Proposition 4. Consider a discrete distribution $\pi = (p(1), \dots, p(k))$ over the state space $S = \{1, \dots, k\}$. In the Metropolis-Hastings MCMC method, given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ defined as

$$A(i, j) = \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \quad (23)$$

satisfy the Equation (16) of detailed balance, that is

$$\frac{T_{ij}}{T_{ji}} = \frac{p(j)}{p(i)}. \quad (24)$$

This means that, any Markov chain simulated for “long enough” by the Metropolis-Hastings method, with the proposal probabilities as defined above, it will generate a random sample from the distribution π .

Proof. (sketch of) We want to prove that

$$\frac{T_{ij}}{T_{ji}} = \frac{Q(i, j)A(i, j)}{Q(j, i)A(j, i)} = \frac{p(j)}{p(i)}. \quad (25)$$

By definition of the acceptance probabilities, we have

$$A(i, j) = \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \quad \text{and} \quad A(j, i) = \min \left\{ 1, \frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)} \right\} \quad (26)$$

We can distinguish two cases:

- $A(i, j) < 1$:

$$\begin{aligned} \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} &< 1 \\ \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} &< 1 \\ \frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)} &> 1 \\ A(j, i) &= 1 \end{aligned} \quad (27)$$

This means that

$$\frac{A(i, j)}{A(j, i)} = \frac{A(i, j)}{1} = \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)}. \quad (28)$$

- $A(i, j) = 1$ implies (it is an “if and only if”) $A(j, i) < 1$. This means that

$$\frac{A(i, j)}{A(j, i)} = \frac{1}{A(j, i)} = \frac{1}{\frac{p(i)}{p(j)} \frac{Q(i, j)}{Q(j, i)}} = \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)}. \quad (29)$$

So, in either case

$$\frac{A(i, j)}{A(j, i)} = \frac{p(j) Q(j, i)}{p(i) Q(i, j)}, \quad (30)$$

from which it follows that

$$\frac{T_{ij}}{T_{ji}} = \frac{Q(i, j)A(i, j)}{Q(j, i)A(j, i)} = \frac{Q(i, j)}{Q(j, i)} \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} = \frac{p(j)}{p(i)}. \quad (31)$$

□

Remark 1. *It is worth to note that*

1. *Since the acceptance probabilities $A(i, j) = \min \left\{ 1, \frac{p(j) Q(j, i)}{p(i) Q(i, j)} \right\}$ in the Metropolis-Hastings MCMC method depend only on the ratios of (discrete) densities, it is sufficient if this ratio can be computed. Also, for the computation of this (density) ratio, the normalization constant does not need to be known.*
2. *To accept a new state with a certain acceptance probability $A(i, j)$, in practice it means that a random number u has to be drawn from a continuous uniform distribution over the interval $[0, 1]$, that is $u \sim U(0, 1)$. Then*
 - *if $u \leq A(i, j)$, the new proposed state is accepted,*
 - *if $u > A(i, j)$, the new proposed state is rejected.*
3. *The Metropolis-Hastings MCMC method can also be applied if the state space S is (countable or uncountable) infinite. Here we have restricted on finite spaces for simplicity. Two examples of MCMC simulations for (i) a countable infinite and (ii) an uncountable infinite state space S are provided in the following.*

Example 4. *(MCMC sampler with countable infinite S)*

Given a Poisson distribution with parameter λ and the discrete density function

$$p(i) = \frac{\lambda^i}{i!} e^{-\lambda}, \quad i = 0, 1, 2, 3, \dots, \quad (32)$$

construct a Markov chain with state space $S = \mathbb{N}_0$, which converges to the distribution

$$\pi = (p(0), p(1), \dots) \quad (33)$$

as stationary distribution.

Solution: For the proposal distribution, we set

$$Q(0, j) = \begin{cases} 1 & j = 1 \\ 0 & \text{otherwise} \end{cases} \quad (34)$$

for $i = 0$, and

$$Q(i, j) = \begin{cases} 0.5 & j = i + 1 \\ 0.5 & j = i - 1 \\ 0 & \text{otherwise} \end{cases} \quad (35)$$

for $i > 0$. Note that such a proposal distribution satisfies the properties listed in Note 6.

Given the proposal probabilities $Q(i, j)$, the acceptance probabilities $A(i, j)$ have to be chosen so that the Equation (16) is satisfied. From Proposition 4, the equation of detailed balance is satisfied by the following acceptance probabilities

$$\begin{aligned} A(i, j) &= \min \left\{ 1, \frac{p(j)}{p(i)} \frac{Q(j, i)}{Q(i, j)} \right\} \\ &= \min \left\{ 1, \frac{\frac{\lambda^j}{j!} e^{-\lambda}}{\frac{\lambda^i}{i!} e^{-\lambda}} HR(i, j) \right\} \\ &= \min \left\{ 1, \lambda^{j-i} \frac{i!}{j!} HR(i, j) \right\} \end{aligned} \quad (36)$$

where

$$HR(i, j) = \frac{Q(j, i)}{Q(i, j)} = \begin{cases} HR(0, 1) = 0.5 \\ HR(1, 0) = 2 \\ HR(i, j) = 1 & \text{otherwise} \end{cases} \quad (37)$$

```
nSim <- 20000
x <- rep(NA, nSim)
x[1] <- 5

lambda <- 5

for (k in 2:nSim) {
  ## propose new state x_new from Q(i,j):
  if(x[k-1] == 0){
    x_new <- 1
  }else{
    p <- rbinom(n = 1, size = 1, prob = .5)
    if(p == 0){
      x_new <- x[k-1] - 1
    }else{
      x_new <- x[k-1] + 1
    }
  }
}

## define Hastings ratio:
if(x[k-1] == 0 & x_new == 1){
  HR <- 0.5
}else{
  if(x[k-1] == 1 & x_new == 2){
    HR <- 2
  }else{
    HR <- 1
  }
}
```

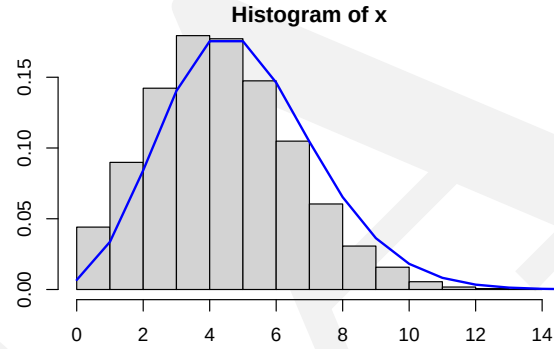
```

## accept or reject x_new according to A(i,j):
Aij <- min(1,
           lambda^(x_new - x[k-1])*factorial(x[k-1])/factorial(x_new)*HR)
u <- runif(n = 1,
          min = 0,
          max = 1)

if(u <= Aij){
  x[k] <- x_new
}else{
  x[k] <- x[k-1]
}
}

hist(x,
     prob = TRUE,
     breaks = sort(unique(x)))
lines(0:20,
      dpois(x = 0:20, lambda = lambda))

```



□

Example 5. (MCMC sampler with uncountable infinite S) Given the standard normal distribution with density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \propto_x e^{-\frac{1}{2}x^2}, \quad (38)$$

construct a Markov chain with state space $S = \mathbb{R}$, which converges to the standard normal distribution as the stationary distribution.

Solution: Given the current state x , we propose a new state y such that

$$y = x + u, \quad u \sim U\left(\left[-\frac{1}{2}, \frac{1}{2}\right]\right). \quad (39)$$

This means that the proposal probabilities are defined as

$$Q(x, y) = \begin{cases} 1 & y \in \left[x - \frac{1}{2}, x + \frac{1}{2}\right] \\ 0 & \text{otherwise} \end{cases} \quad (40)$$

Note that such a proposal distribution satisfies the properties listed in Note 6.

Given the proposal probabilities $Q(x, y)$, the acceptance probabilities $A(x, y)$ have to be chosen so that the Equation (16) is satisfied. From Proposition 4, the equation of detailed balance is satisfied by the following acceptance probabilities

$$\begin{aligned} A(x, y) &= \min \left\{ 1, \frac{p(y) Q(y, x)}{p(x) Q(x, y)} \right\} \\ &= \min \left\{ 1, \frac{e^{-\frac{1}{2}y^2} 1}{e^{-\frac{1}{2}x^2} 1} \right\} \\ &= \min \left\{ 1, e^{\frac{1}{2}(x^2 - y^2)} \right\}. \end{aligned} \tag{41}$$

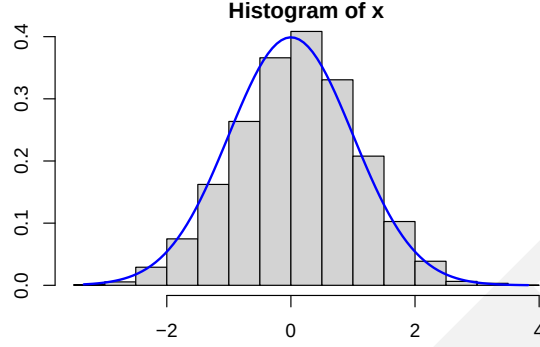
```
nSim <- 20000
x <- rep(NA, nSim)
x[1] <- 0

for (k in 2:nSim) {
  ## propose new state x_new from Q(i,j):
  x_new <- x[k-1] + runif(1, -0.5, 0.5)

  ## accept or reject x_new according to A(i,j):
  Aij <- min(1, exp(0.5*(x[k-1]^2 - x_new^2)))
  u <- runif(n = 1,
             min = 0,
             max = 1)

  if(u <= Aij){
    x[k] <- x_new
  }else{
    x[k] <- x[k-1]
  }
}

pdf("./standard-normal-histogram.pdf", width = 5, height = 3)
par(mar = c(2,2,1,1))
hist(x,
      prob = TRUE)
lines(seq(min(x),
          max(x),
          length.out = 100),
      dnorm(x = seq(min(x),
                    max(x),
                    length.out = 100),
            mean = 0,
            sd = 1),
      col = 'blue',
      lwd = 2)
dev.off()
```



□

3 Bayesian inference

In Bayesian statistics, the unknown parameters θ are considered as random variable following certain **prior distributions** that are imposed on them. The prior(s) should reflect the pre-knowledge (prior belief) about the parameters θ . The prior distribution has a density $p(\theta)$. Then, given observed data $\mathbf{y} = (y_1, \dots, y_n)'$, the density $p(\mathbf{y}|\theta)$ is the **likelihood**, describing how “likely” the data $\mathbf{y}$ would be if θ was the true vector parameter. The goal of a Bayesian analysis is to compute the **posterior distribution** $p(\theta|\mathbf{y})$, or at least to generate some samples from it, so as to approximate its properties/features. From Bayes theorem, we know that

$$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})}. \quad (42)$$

Furthermore, since the marginal likelihood $p(\mathbf{y})$ does not depend on θ , the posterior density is proportional, as a function of only the parameters θ , to the likelihood times the prior density, that is

$$p(\theta|\mathbf{y}) \propto_{\theta} p(\mathbf{y}|\theta)p(\theta). \quad (43)$$

A prior is called **conjugate** if the posterior distribution falls in the same family type as the prior distribution. If the posterior distribution density has not the shape of the density of any known distribution, Markov Chain Monte Carlo (MCMC) simulations are used to generate samples from the posterior distribution. In the next subsection, a short review of MCMC simulations is given, for the case of uncountable infinite state space for a univariate random variable θ . Then, the Metropolis-Hastings MCMC method is introduced for both the univariate and multivariate parameter spaces.

3.1 Markov Chain Monte Carlo (MCMC) - univariate case

To infer a parameter vector θ under a Bayesian setting, the key idea is to construct a Markov Chain whose stationary distribution is the posterior distribution $p(\theta|\mathbf{y})$. Subsequently, a realization of this Markov chain is generated. As the Markov chain converges to its stationary distribution (in this case, to the posterior distribution), the realization (after having removed the initial samples called **burn-in phase**) will be a posterior sample. This posterior sample can then be used to approximate all the quantities of interest of the posterior distribution.

For simplicity, we will start focusing on the case when the parameter space is univariate, that is the vector parameter $\boldsymbol{\theta}$ reduces to a scalar θ . An MCMC simulation starts with the initialization of the unknown parameter, e.g. $\theta = \theta^{(1)}$. Then, the subsequent iterations are performed within a for-loop, defined as follows.

Each MCMC iteration $t \in \{1, \dots, 2T\}$ consists of two steps:

1. It is proposed to move from the current value $\theta^{(t)}$ of the unknown parameter to a new value θ^* . Often, this new candidate value is obtained by adding a small random number v (with expectation 0) to the current value $\theta^{(t)}$, for example

$$\theta^* = \theta^{(t)} + v, \quad v \sim U([- \epsilon, \epsilon]). \quad (44)$$

In general $\theta^* \sim Q(\theta^{(t)}, \theta^*)$, where Q is the proposal distribution.

2. The new value is either accepted or rejected according to some acceptance probability $A(\theta^{(t)}, \theta^*)$. If accepted, the next value is set to the candidate value, that is $\theta^{(t+1)} = \theta^*$. If rejected, the current value is left unchanged, that is $\theta^{(t+1)} = \theta^{(t)}$.

3.2 Metropolis-Hastings Algorithm - univariate case

In the Metropolis-Hastings MCMC method (univariate case), the previous algorithm becomes as follows:

1. **(Initialization):** set $\theta^{(1)} = \theta \in \Theta$, where $\theta^{(1)}$ can be any possible value from the parameter space Θ .
2. **(MCMC iterations):** for $t = 1, \dots, 2T$
 - (a) Propose to move from the current state $\theta^{(t)}$ to θ^* , where the candidate value θ^* is proposed with proposal probability $Q(\theta^{(t)}, \theta^*)$.
 - (b) Accept the new state θ^* with acceptance probability $A(\theta^{(t)}, \theta^*)$, that is
 - compute $A(\theta^{(t)}, \theta^*)$ consistent with Eq. (23);
 - draw a random number $u \sim U([0, 1])$
 - if $u \leq A(\theta^{(t)}, \theta^*)$, then set $\theta^{(t+1)} = \theta^*$ (accept the candidate)
 - if $u > A(\theta^{(t)}, \theta^*)$, then set $\theta^{(t+1)} = \theta^{(t)}$ (leave the state unchanged)
 - (c) Do the next iteration.

The proposal probabilities $Q(\theta^{(t)}, \theta^*)$

In the Metropolis-Hastings MCMC algorithm, the proposal probabilities $Q(\theta^{(t)}, \theta^*)$ in step 2(a) can be chosen arbitrarily, as long as two conditions are fulfilled:

1. $Q(x, y) > 0$ if and only if $Q(y, x) > 0$.
2. Every parameter $x \in \Theta$ must be reached from every parameter $y \in \Theta$, in a finite number of steps.

A standard approach was already mentioned in the previous section: given $\theta^{(t)}$, a random number v is sampled from a uniform distribution over the interval $[-\epsilon, \epsilon]$, that is $v \sim U([- \epsilon, \epsilon])$, and the new candidate value is defined as

$$\theta^* = \theta^{(t)} + v. \quad (45)$$

The acceptance probabilities $A(\theta^{(t)}, \theta^*)$

In the Metropolis-Hastings MCMC algorithm, the acceptance probabilities $A(\theta^{(t)}, \theta^*)$ in step 2(b) have to be computed consistent with Eq.(23), that is

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\theta^*|\mathbf{y})}{p(\theta^{(t)}|\mathbf{y})} \frac{Q(\theta^*, \theta^{(t)})}{Q(\theta^{(t)}, \theta^*)} \right\} \quad (46)$$

As we have seen in Proposition 4, this choice of $A(\theta^{(t)}, \theta^*)$ ensures that the target distribution, in this case the posterior distribution, fulfills the Equation of Detailed Balance (see Eq. (16)). This means that such a defined MCMC if simulated for “long enough” will “converge” to the posterior distribution $p(\theta|\mathbf{y})$ as its stationary distribution. Moreover, for the proposal mechanism described above, the Hastings Ratio $\frac{Q(\theta^*, \theta^{(t)})}{Q(\theta^{(t)}, \theta^*)}$ is always equal to 1. Therefore, the acceptance probability reduces to

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\theta^*|\mathbf{y})}{p(\theta^{(t)}|\mathbf{y})} \right\} \quad (47)$$

Furthermore, since

$$p(\theta|\mathbf{y}) \propto_\theta p(\mathbf{y}|\theta)p(\theta), \quad (48)$$

the acceptance probability further reduces to

$$A(\theta^{(t)}, \theta^*) = \min \left\{ 1, \frac{p(\mathbf{y}|\theta^*)p(\theta^*)}{p(\mathbf{y}|\theta^{(t)})p(\theta^{(t)})} \right\}. \quad (49)$$

3.3 Metropolis-Hastings Algorithm - multivariate case

In more realistic scenarios it is likely that the mathematical model under investigation has a multivariate parameter space Θ of dimension K . In this case, the univariate Metropolis-Hastings MCMC method can be extended as follows. Assume that the underlying model has an unknown K -dimensional vector parameter

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_K). \quad (50)$$

Then every MCMC iteration consists of K substeps. That is, in iteration t we move from the current state

$$(\theta_1^{(t)}, \dots, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \quad (51)$$

to the new state

$$(\theta_1^{(t+1)}, \dots, \theta_k^{(t+1)}, \theta_{k+1}^{(t+1)}, \dots, \theta_K^{(t+1)}) \quad (52)$$

in K successive substeps:

- (step 1): Propose to go from $(\theta_1^{(t)}, \theta_2^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^*, \theta_2^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$
- (step 2): Propose to go from $(\theta_1^{(t+1)}, \theta_2^{(t)}, \theta_3^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \theta_2^*, \theta_3^{(t)}, \dots, \theta_k^{(t)}, \dots, \theta_K^{(t)})$
- $\vdots$
- (step k): Propose to go from $(\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)})$
- $\vdots$

- (step K): Propose to go from $(\theta_1^{(t+1)}, \dots, \theta_{K-1}^{(t+1)}, \theta_K^{(t)})$ to $(\theta_1^{(t+1)}, \dots, \theta_{K-1}^{(t+1)}, \theta_K^*)$

where

$$\theta_k^{(t+1)} = \begin{cases} \theta_k^{(t)} & \text{if the } k\text{-th move is rejected} \\ \theta_k^* & \text{if the } k\text{-th move is accepted} \end{cases} \quad (53)$$

That is, the k -th step proposes to move from $\boldsymbol{\theta}_A$ to $\boldsymbol{\theta}_B$, where

$$\begin{aligned} \boldsymbol{\theta}_A &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \\ \boldsymbol{\theta}_B &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}), \end{aligned} \quad (54)$$

whose acceptance probability is

$$A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) = \min \left\{ 1, \frac{p(\mathbf{y}|\boldsymbol{\theta}_B)p(\boldsymbol{\theta}_B)}{p(\mathbf{y}|\boldsymbol{\theta}_A)p(\boldsymbol{\theta}_A)} \right\}. \quad (55)$$

After $2T$ steps, the algorithm defined above will generate a posterior sample of the form

$$\begin{aligned} \boldsymbol{\theta}^{(1)} &= (\theta_1^{(1)}, \dots, \theta_K^{(1)}) \\ \boldsymbol{\theta}^{(2)} &= (\theta_1^{(2)}, \dots, \theta_K^{(2)}) \\ &\vdots \\ \boldsymbol{\theta}^{(2T)} &= (\theta_1^{(2T)}, \dots, \theta_K^{(2T)}). \end{aligned} \quad (56)$$

3.4 Gibbs sampling

When the **full conditional distribution** (fcd) of the k -th parameter θ_k can be computed, that is it has an explicit analytical form, then the k -th substep of the Metropolis-Hastings algorithm can be replaced by a more efficient Gibbs sampling step. The density of the fcd of the k -th parameter is defined as

$$p(\theta_k | \theta_1, \dots, \theta_{k-1}, \theta_{k+1}, \dots, \theta_K, \mathbf{y}) \quad (57)$$

or short

$$p(\theta_k | \boldsymbol{\theta}_{-k}, \mathbf{y}), \quad (58)$$

where $\boldsymbol{\theta}_{-k}$ denotes all parameters except for the k -th component. In the k -th substep of the Gibbs MCMC algorithm, instead of the Metropolis-Hastings move

$$\theta_k^* = \theta_k^{(t)} + v, \quad v \sim U([- \epsilon, \epsilon]), \quad (59)$$

we re-sample θ_k^* from its full conditional distribution. In this case, the acceptance probability $A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)$ of the k -th submove is equal to 1. In fact, recall that the k -th substep proposes to move from $\boldsymbol{\theta}_A$ to $\boldsymbol{\theta}_B$, where

$$\begin{aligned} \boldsymbol{\theta}_A &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^{(t)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \\ \boldsymbol{\theta}_B &= (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_k^*, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}). \end{aligned} \quad (60)$$

Then, if we define

$$\boldsymbol{\theta}_{-k}^{(t)} := (\theta_1^{(t+1)}, \dots, \theta_{k-1}^{(t+1)}, \theta_{k+1}^{(t)}, \dots, \theta_K^{(t)}) \quad (61)$$

as the vector of the current states except the k -th component, the acceptance probability for the k -th step is

$$A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) = \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B|\mathbf{y})}{p(\boldsymbol{\theta}_A|\mathbf{y})} \frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} \right\}, \quad (62)$$

where the Hastings ratio reduces to

$$\begin{aligned} \frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} &= \frac{p(\theta_k^{(t)}|\boldsymbol{\theta}_{-k}^{(t)}, \mathbf{y})}{p(\theta_k^*|\boldsymbol{\theta}_{-k}^{(t)}, \mathbf{y})} \\ &= \frac{\frac{p(\theta_k^{(t)}, \boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})}{p(\boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})}}{\frac{p(\theta_k^*, \boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})}{p(\boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})}} \\ &= \frac{p(\theta_k^{(t)}, \boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})}{p(\theta_k^*, \boldsymbol{\theta}_{-k}^{(t)}|\mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A, \mathbf{y})}{p(\boldsymbol{\theta}_B, \mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A|\mathbf{y})p(\mathbf{y})}{p(\boldsymbol{\theta}_B|\mathbf{y})p(\mathbf{y})} \\ &= \frac{p(\boldsymbol{\theta}_A|\mathbf{y})}{p(\boldsymbol{\theta}_B|\mathbf{y})}. \end{aligned} \quad (63)$$

Therefore,

$$\begin{aligned} A(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B) &= \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B|\mathbf{y})}{p(\boldsymbol{\theta}_A|\mathbf{y})} \frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)} \right\} \\ &= \min \left\{ 1, \frac{p(\boldsymbol{\theta}_B|\mathbf{y})}{p(\boldsymbol{\theta}_A|\mathbf{y})} \frac{p(\boldsymbol{\theta}_A|\mathbf{y})}{p(\boldsymbol{\theta}_B|\mathbf{y})} \right\} \\ &= 1. \end{aligned} \quad (64)$$

Remark 2. Instead of $\frac{Q(\boldsymbol{\theta}_B, \boldsymbol{\theta}_A)}{Q(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B)}$ we could simply write $\frac{Q(\theta_k^*, \theta_k^{(t)})}{Q(\theta_k^{(t)}, \theta_k^*)}$, as only the k -th parameter changes. Furthermore, since the new state of the k -th parameter is sampled from its full conditional distribution, it does not even depend on its current state. This means that the Hastings ratio can be further written as $\frac{Q(\theta_k^{(t)})}{Q(\theta_k^*)}$.

3.5 Example

3.6 Bayesian Uncertainty Quantification

Assume that we want to estimate the parameters $\boldsymbol{\theta}$ of a given model $\mathcal{M}_{\boldsymbol{\theta}}$, based on observed experimental data. As mentioned earlier, in a Bayesian framework first a prior distribution $p(\boldsymbol{\theta})$ is imposed on the vector parameter $\boldsymbol{\theta}$. Then, given some data $\mathbf{y}$ and a likelihood function $p(\mathbf{y}|\boldsymbol{\theta})$ (typically based on a “distance” between observed data and the corresponding model predictions), we can update our knowledge about $\boldsymbol{\theta}$ after having observed data by computing the posterior

distribution

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto_{\boldsymbol{\theta}} p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}). \quad (65)$$

If we knew the exact analytical form of the posterior, we could have draw conclusion about some quantities of interest regarding $\boldsymbol{\theta}$, such as the mean, its variance, and confidence intervals/regions. However, often the posterior $p(\boldsymbol{\theta}|\mathbf{y})$ is unknown but can be approximated by MCMC samples. In fact, once such samples are generated, for example via the Metropolis-Hastings MCMC method, the properties of interest of the posterior distribution can be empirically approximated using sample statistics (e.g. the sample mean, sample median, sample variance, confidence intervals, etc.).

It is also worth to note that once a MCMC sample $\{\boldsymbol{\theta}^{(t)}\}_t$ is generated from the multivariate posterior distribution $p(\boldsymbol{\theta}|\mathbf{y})$, each marginal MCMC sample $\{\theta_j^{(t)}\}_t$ can be used to approximate properties of the posterior distribution $p(\theta_j|\mathbf{y})$ of the j -th component θ_j of $\boldsymbol{\theta}$. In the following, we present two techniques to build confidence intervals for each component θ_j of $\boldsymbol{\theta}$, based on their posterior distributions $p(\theta_j|\mathbf{y})$.

Quantile-based confidence intervals

Perhaps the easiest way to obtain a confidence interval of a scalar parameter θ is to use the quantiles of its posterior distribution. At a level α , to make a $100 \times (1 - \alpha)\%$ quantile-based confidence interval for θ , it is required to find the $\alpha/2$ and $1 - \alpha/2$ posterior quantiles of θ , that is the numbers $\theta_{\alpha/2} < \theta_{1-\alpha/2}$ satisfying

$$\begin{aligned} P(\theta < \theta_{\alpha/2}|\mathbf{y}) &= \alpha/2 \\ P(\theta > \theta_{1-\alpha/2}|\mathbf{y}) &= \alpha/2. \end{aligned} \quad (66)$$

This means that $\theta_{\alpha/2}$ and $\theta_{1-\alpha/2}$ also satisfy

$$\begin{aligned} P(\theta \in [\theta_{\alpha/2}, \theta_{1-\alpha/2}]|\mathbf{y}) &= 1 - P(\theta \notin [\theta_{\alpha/2}, \theta_{1-\alpha/2}]|\mathbf{y}) \\ &= 1 - [P(\theta < \theta_{\alpha/2}|\mathbf{y}) + P(\theta > \theta_{1-\alpha/2}|\mathbf{y})] \\ &= 1 - [\alpha/2 + \alpha/2] = 1 - \alpha. \end{aligned} \quad (67)$$

Therefore, $\theta_{\alpha/2}$ and $\theta_{1-\alpha/2}$ define the lower and upper bounds of a $100 \times (1 - \alpha)\%$ quantile-based confidence interval for θ . One limitation of this method is that there might be θ values outside the quantile-based interval that have a higher posterior probability density than some points inside the interval, as shown in the example of Fig. 1.

Highest posterior density region

We have seen earlier that a quantile-based confidence interval can contain θ values with lower posterior probability density than some value outside the confidence interval. This limitation can be overcome by a more restrictive type of confidence interval.

Definition 7. Given a level α , a $100 \times (1 - \alpha)$ **highest posterior density (HPD)** region for a parameter $\theta \in \Theta$, consists of a subset $S_{\mathbf{y}} \subset \Theta$ of the parameter space Θ , such that

1. $P(\theta \in S_{\mathbf{y}}|\mathbf{y}) = 1 - \alpha$
2. if $\theta_a \in S_{\mathbf{y}}$ and $\theta_b \notin S_{\mathbf{y}}$, then $P(\theta_a|\mathbf{y}) > P(\theta_b|\mathbf{y})$.

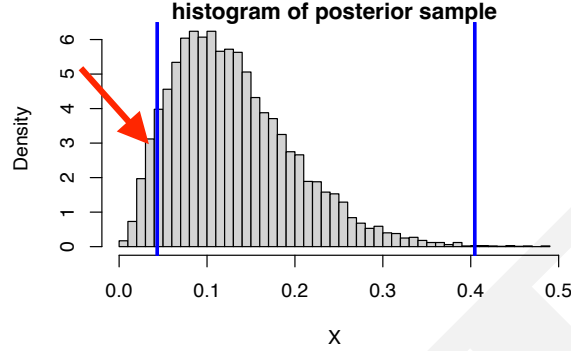


Figure 1: An illustrative example of histogram of a MCMC posterior sample of θ . The vertical blue lines indicate the lower and upper bounds of a 5% quantile-based confidence interval. The red arrow points to a θ value outside the confidence interval that has a higher posterior probability density than some points inside the interval.

Therefore, all points in an HPD region have a higher posterior density than points outside the region. However, an HPD region might not be an interval if the posterior density is multimodal (that is, having multiple peaks). This is why we refer to it as region instead of as an interval.

An HPD region can be constructed by

- Gradually move a horizontal line down across the density, including in the HPD region all the θ values having a density above the horizontal line.
- Stop moving the line down when the posterior probability of the θ values in the HPD region reaches $(1 - \alpha)$.

An HPD region is computed for the posterior sample of previous example, and the result is displayed in Fig. 2, where a comparison is made with the quantile-based confidence interval previously computed.

3.7 Posterior Predictive Inference

After obtaining an MCMC sample from the posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{y}), \quad (68)$$

where $\boldsymbol{\theta}$ denotes the model parameters and $\mathbf{y}$ the observed data, we can make predictions about new or unobserved data points $\tilde{y}$. The posterior predictive distribution is defined as

$$p(\tilde{y} \mid \mathbf{y}) = \int p(\tilde{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta}. \quad (69)$$

This integral represents an average of the likelihood of $\tilde{y}$ over the posterior uncertainty in $\boldsymbol{\theta}$. Computing it analytically is often intractable, but MCMC samples $\{\boldsymbol{\theta}^{(s)}\}_{s=1}^S$ from the posterior provide a natural Monte Carlo approximation. Specifically, by the law of large numbers, if the

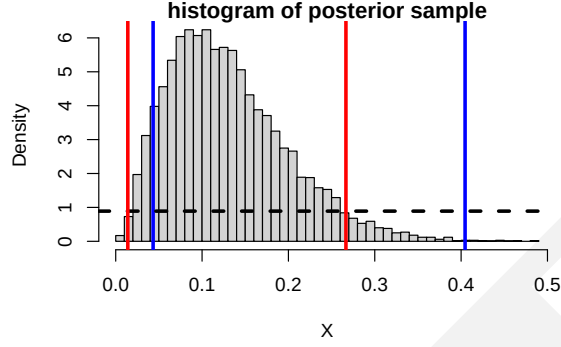


Figure 2: An illustrative example of histogram of a MCMC posterior sample of θ . The vertical blue lines indicate the lower and upper bounds of a 5% quantile-based confidence interval. The vertical red lines indicate the lower and upper bounds of a 5% HPD region, identified by the horizontal dashed black line.

draws $\theta^{(s)}$ are (approximately) independent samples from $p(\theta | \mathbf{y})$, we have

$$\int p(\tilde{y} | \theta) p(\theta | \mathbf{y}) d\theta \approx \frac{1}{S} \sum_{s=1}^S p(\tilde{y} | \theta^{(s)}). \quad (70)$$

The equation above approximates the density at a fixed $\tilde{y}$. However, in practice, we often want actual predictive samples rather than the density itself. In other words, the two-step procedure defined by

1. Each $\theta^{(s)}$ is drawn from $p(\theta | \mathbf{y})$,
2. Conditional on $\theta^{(s)}$, the new data point $\tilde{y}$ has the distribution $p(\tilde{y} | \theta)$,

is equivalent to drawing $\tilde{y}$ from the mixture distribution defined by the integral of Eq. (69), or its Monte Carlo approximation given by Eq. (70).

Intuitively, each posterior draw $\theta^{(s)}$ represents a plausible parameter configuration given the observed data. Averaging over these draws approximates integrating over all plausible parameter values weighted by their posterior probability. In practice, to generate a posterior predictive sample, for each MCMC draw $\theta^{(s)}$, we simulate

$$\tilde{y}^{(s)} \sim p(\tilde{y} | \theta^{(s)}). \quad (71)$$

The collection of simulated values $\{\tilde{y}^{(s)}\}_{s=1}^S$ provides an empirical approximation of the posterior predictive distribution. Prediction intervals for $\tilde{y}$ can then be obtained by taking quantiles of this sample. For example, a 95% predictive interval is given by the 2.5th and 97.5th percentiles of $\{\tilde{y}^{(s)}\}$. This approach naturally accounts for both parameter uncertainty (through the posterior $p(\theta | \mathbf{y})$) and inherent randomness in the data (through $p(\tilde{y} | \theta)$), giving a coherent Bayesian measure of uncertainty for future observations.